THE CONORMAL IDENTITY FOR A PAIR OF IDEALS GENERATED BY AN H1-REGULAR SEQUENCE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- H_1-regularity of a sequence and its use in computing intersections of ideals. the statement counted by the sibling card S1-algebra-H1-regular-in-quotient at 7471-7494, cited in the proof at proof lines 7515-7518. Supplies the step that turns the middle part of the decomposition into a product.
- Generators of a quotient of ideals and the resulting decomposition of an element of the square. used at proof lines 7503-7512. Produces the three-part form of an element of the intersection.
- The product ideal IJ and the elementary identities IJ contained in I cap J and I^2 contained in IJ when I is contained in J. used at proof lines 7503-7519 to recognise that all three parts lie in IJ. Supplies the conclusion after the middle part is dealt with.

Source reference. Original lemma 0666: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7925–7925. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0665: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7924–7924. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a ring. Let $I \subset J \subset A$ be ideals. Assume that $J/I \subset A/I$ is generated by an H_1-regular sequence. Then $I \cap J^2 = IJ$.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. To prove this choose $g_1, \ldots, g_m \in J$ whose images in A/I form a H_1-regular sequence which generates J/I. In particular $J = I + (g_1, \ldots, g_m)$. Suppose that $x \in I \cap J^2$. Because $x \in J^2$ can write

$$x = \sum a_{ij} g_i g_j + \sum a_j g_j + a$$

with $a_{ij} \in A$, $a_j \in I$ and $a \in I^2$. Then $\sum a_{ij} g_i g_j \in I \cap (g_1, \ldots, g_m)$ hence by Lemma 0665 we see that $\sum a_{ij} g_i g_j \in I(g_1, \ldots, g_m)$. Thus $x \in IJ$ as desired. $\square$